Qualifying Exam in Microeconomics
Arizona State University
August 2013

There are five questions and you have four hours to complete the exam.

1. A risk-neutral monopolist insurer sells to a single, risk averse consumer. The consumer has initial wealth of $w > 0$ and faces a potential loss of size L with probability p, where $0 < L < w$ and $0 < p < 1$. The consumer has expected utility preferences represented by a strictly concave vN-M utility u, which is C^1 with $u' > 0$ on its domain of $\mathbb{R}_+$. The insurer offers a contract $(x, t) \geq 0$ consisting of a transfer t that the consumer pays to insurer in *both* states of the world and an indemnity payment x that the insurer pays to the consumer if the loss occurs. The insurer's expected cost of a contract (x, t) is $c(x) = \gamma px$, where $\gamma > 1$. The insurer chooses a contract to maximize expected profit, $t - \gamma px$, subject to the constraint that the consumer weakly prefers the contract offered to the no-trade contract, $(0, 0)$. The insurer knows w, L, p, γ, and $u(\cdot)$.

   (a) Does it follow from the assumptions made that the insurer offers the consumer *less than* full insurance ($x < L$)? If so, prove it. If not, give additional sufficient conditions for the conclusion and explain why your conditions are sufficient.

   (b) For $\tau = 0, 1$, let u_τ be a vN-M utility satisfying the assumptions of this problem, with u_1 more risk averse than u_0 (in the sense of Arrow-Pratt). For $\tau = 0, 1$, let V_τ be the monopolist's maximum expected profit when the consumer's vN-M utility is u_τ. What can you say about the sign of $V_1 - V_0$? Prove it.

   (c) Using the notation from part (b), for $\tau = 0, 1$ let γ_τ be the *smallest* value of γ such that the no-trade contract $(0, 0)$ maximizes the insurer's expected profit when the consumer's vN-M utility is u_τ. What can you say about the sign of $\gamma_1 - \gamma_0$? How do you know?

2. The following prisoners' dilemma game

		Player i+1	
		C	D
Player i	C	2 2	−1 3
	D	3 −1	0 0

is played by a countably infinite population of players, $\{1, 2, 3 \ldots i, \ldots\}$ whose generations overlap. At date 1, players 1 and 2 play the game; at date 2, players 2 and 3 play the game. More generally, at date t, players t and $t+1$ play the game. Notice that player 1 only plays once; every other player plays twice, once when young, once when old; and no one plays the same person more than once. Players evaluate outcomes of the game by the present discounted value of the payoffs they get for the (at most two) periods they play; the players have a common discount factor of $\delta \in (0, 1)$. (For example if in some outcome the payoff of player $i \geq 2$ is π_i^{i-1} at date $i-1$ and π_i^i at date i, then i's utility of the game's outcome is $\pi_i^{i-1} + \delta\pi_i^i$.) At each date t, each player observes the past history of play through date $t-1$.

(Question 2 continued) You are asked to evaluate the following assertion:

> There a $\delta^* \in (0,1)$ such that, if $\delta \in (\delta^*, 1)$, then there is a subgame-perfect Nash equilibrium in which C is sometimes chosen in the equilibrium history.

If it is true, prove it: in particular, identify a strategy profile that supports such an outcome, explain clearly why it is subgame perfect, and identify δ^* for your strategy profile. If it is false, prove that, for every $\delta \in (0,1)$, the only SPNE outcome is "DD at every date," and clearly explain how you use subgame perfection.

3. A consumer's preferences for consumption plans in $X = \mathbb{R}^L_+$ are described by a *complete, transitive, continuous,* and *strongly monotone* binary relation $\succsim \in X \times X$. (Note that convexity of $\succsim$ is *not* assumed.) The demand correspondence is

$$d(p, w) = \{y \in B(p, w) \mid y \succsim x \text{ for every } x \in B(p, w)\}$$

where $B(p, w) = \{x \in X \mid p \cdot x \leq w\}$ is the budget correspondence, with p the price vector and w the consumer's wealth. The domain of each correspondence is $\mathbb{R}^{L+1}_{++}$. The consumer's wealth is equal to the value of the endowment, $p \cdot \omega$, where $\omega \in \mathbb{R}^L_{++}$. In what follows, for a point $y \in \mathbb{R}^n$, let $y_{-1} = (y_2, ..., y_n)$, that is, the vector y with the first component deleted. Let $p^0 >> 0$ and $p^1 = (p^1_1, p^0_{-1})$, where $0 < p^1_1 < p^0_1$; and for $t = 0, 1$, let

$$x^t \in d(p^t, p^t \cdot \omega),$$

where $x^0_1 > \omega_1$. In words, the consumer is a *net buyer* of good 1 at p^0 and the price of good 1 *falls* between p^0 and p^1.

Does it follow that $x^1_1 > \omega_1$? If so *prove* it, if not explain *clearly* how the conclusion can fail.

4. Two firms compete, via Bertrand competition, for a worker of unknown productivity. They share a common prior of 1/2 that the worker is productive (high type), in which case marginal product equals 2. With complementary probability he is unproductive (low type), in which case marginal product equals 0. Assume that workers can send a signal to the firms by undertaking a costly task, e.g. education. Task choice takes values in the interval $[0, \infty)$. The high type's cost function from undertaking level t of the task is $c_H(t) = t^2$, and the low type's cost function is $c_L(t) = 2t^2$. Assume worker utility from a pair (w, t) is $u_H(w, t) = w - c_H(t)$, and similarly for the low type. Finally, lotteries on wages are evaluated using expected utility. Fixing this environment, consider the following two extensive form games.

   A. (Signaling) At time 0, the worker undertakes a task choice t. Both firms observe this level t and form a common belief about the worker's productivity. At time 1, firms offer competing wages (a la Bertrand) for the worker. Assume the worker chooses the firm who offers a higher wage and breaks ties by toss of fair coin.

   B. (Screening) At time 0, firms offer competing menus of contracts (a la Bertrand, they make offers of menus simultaneously), i.e. a menu of contracts is a set of pairs of the form (w, t) where w denotes a wage and t denotes a task level. At time 1 a worker chooses a firm and a pair (w, t) from the menu that maximizes $u(w, t)$. If utility from menus is the same at both firm offers, the tie is resolved by toss of a fair coin. If a worker chooses pair (w, t) he is contractually bound (i.e. under utility penalty of $-\infty$) to complete task level t.

(Question 4 continued) Answer the following questions about these two games:

(a) For the first game, find bounds, $\underline{t}, \bar{t}$, such that (i) for any level $t \in [\underline{t}, \bar{t}]$ there is a separating (sequential) equilibrium in which high types choose t and low types choose 0, and (ii) for any $t \notin [\underline{t}, \bar{t}]$ there is no separating equilibrium. For any putative equilibrium t, you must specify (i) a pair (σ, μ) consisting of strategies and beliefs, and (ii) verify that these pairs satisfy the requisite properties of sequential equilibrium.

(b) For the second game, show that there can be at most one (subgame perfect) separating equilibrium outcome and find this outcome. Here by "outcome" we mean the (not necessarily distinct) pairs (w_H, t_H), (w_L, t_L) chosen by (resp.) the high (low) type. For this part, you may assume that – in any SPNE – both firms are earning zero profit.[1]

5. A risk-neutral principal contracts with a risk-neutral agent to complete a task which entails costly effort. Output takes two values $x \in \{x_H = 5, x_L = 10\}$ and effort can be high or low with $e \in \{e_L = 0, e_H = 1\}$. The agent's effort choice is unobservable to the principal but the distribution of output is a function of effort choice, given as follows:

	x_H	x_L
$f(\cdot\|e_H)$	.8	.2
$f(\cdot\|e_L)$	.5	.5

Assume the following timing structure. At time 0, the principal offers a take-it-or-leave-it contract, consisting of a pair (w_H, w_L), where w_H (resp. w_L) specifies a wage to be paid when x_H (resp. x_L) is realized. If the agent accepts, then at time 1 he undertakes a (private) choice of effort e. At time 2, output is realized and (if he accepts the contract) payoffs are $u(w, e) = f(x_H|e) \cdot w_H + f(x_L|e)w_L - c(e)$ (for the agent) and $f(x_H|e)(x_H - w_H) + f(x_L|e)(x_L - w_L)$ (for the principal), where we assume $c(e_H) = 1, c(e_L) = 0$. If the agent rejects the time 0 offer, he obtains an outside option valued at 0. Assume that the agent accepts the contract whenever indifferent. Answer the following questions:

(a) Find an optimal contract (w_H, w_L) and compute the principal's profit at the optimum. Is the optimal contract unique? If you say it is, prove uniqueness. If not, compute the set of optimal contracts.

(b) Now assume there is an additional *limited liability* constraint, where we require $w_H, w_L \geq 0$. Compute the optimal contract within the class of pairs (w_H, w_L) that satisfy this constraint.

[1] Since cost functions are concretely specified, argue directly as opposed to drawing graphs.